

Highlights

Deep Learning Classification of Cardiac Auscultation Sounds: A Data-Centric 33-Class Framework via Dynamic Noise Augmentation and Sim-to-Real Domain Adaptation

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- Classifies 33 cardiac sound categories using a unified ResNet18-MLP architecture (11.5M parameters).
- Demonstrates structural feature resilience under dynamic hospital noise augmentation ($\mathcal{U}(-5, 5)$ dB).
- Quantifies a severe sim-to-real domain gap, with zero-shot accuracy dropping to 15.7% on human subjects.
- Mitigates the domain shift via head-only fine-tuning, recovering human "Normal" baseline accuracy to 98.6%.
- Transparently validates that adaptation is a decision boundary expansion, causing a 73.6% to 69.9% sim F1-macro trade-off.

Deep Learning Classification of Cardiac Auscultation Sounds: A Data-Centric 33-Class Framework via Dynamic Noise Augmentation and Sim-to-Real Domain Adaptation

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ABSTRACT

Cardiovascular diseases remain the leading cause of mortality worldwide, making early detection through cardiac auscultation critical yet dependent on clinician expertise. Although machine learning shows promise for binary heart sound classification, multi-class pathology identification in high-noise environments remains challenging. This study introduces a two-phase, data-centric framework for high-granularity classification (33 distinct categories), designed as an *in silico* feasibility study to investigate sim-to-real adaptation mechanisms. In Phase 1, we established a robust baseline using 2,010 samples from a high-fidelity auscultation manikin. Signals underwent Mel-spectrogram extraction coupled with dynamic hospital noise augmentation ($U(-5, 5)$ dB SNR). A unified ResNet18-MLP architecture (11.5M parameters)—utilizing a rigid backbone for feature extraction and a dense 280K parameter head for plasticity—achieved a 73.6% independent SAM test F1-macro score. In Phase 2, we directly quantified the simulator-to-reality domain gap. Deployed zero-shot on real-world smartphone recordings from healthy human subjects, the model exhibited severe domain shift, predicting a "Normal" rhythm with only 15.7% accuracy due to human thoracic acoustics mathematically mimicking pathological spectral signatures. We demonstrate that head-only fine-tuning (Leave-One-Subject-Out protocol) mitigates this gap, recovering human baseline accuracy to 98.6%. Crucially, we transparently reveal that this high adaptation is fundamentally an expansion of the "Normal" decision boundary, which subsequently swallows adjacent simulated pathologies and causes a controlled degradation of the simulated F1-macro score to 69.9%. This work exposes the critical limitations of purely simulated training and establishes a transparent proof-of-concept for adaptation, emphasizing that clinical deployment fundamentally requires future fine-tuning on pathological human cohorts.

1. Introduction

Cardiovascular diseases (CVDs), especially ischemic heart disease and stroke, are the leading cause of death and disability worldwide and a major barrier to global development (Mensah et al., 2019). Global CVD prevalence has more than doubled in recent decades, from about 271 million cases in 1990 to over 523 million in 2019, and CVD deaths rose from 12.1 to 18.6 million in the same period (Roth et al., 2017). Updated estimates show this trend continuing, with around 626 million prevalent CVD cases and 19.2 million deaths in 2023, and roughly four in five CVD disability-adjusted life years attributable to modifiable risk factors such as high blood pressure, unhealthy diets, high LDL cholesterol, obesity, diabetes, air pollution, smoking, and physical inactivity (Tan et al., 2025). Projections to 2050 anticipate a ~90% rise in crude CVD prevalence and more than 35 million annual CVD deaths, driven largely by population growth and ageing despite improvements in age-standardized mortality (Chong et al., 2025).

In this context of a sustained and unevenly distributed global CVD burden, heaviest in low- and lower-middle income regions, there is a pressing need for accessible, low-cost, front-line diagnostic strategies to support earlier detection and triage (GBD Collaborators, 2025). Cardiac auscultation, introduced two centuries ago with Laënnec's stethoscope, remains an inexpensive and widely available technique that can detect many structural heart abnormalities when performed and interpreted correctly; it provides crucial screening and prognostic information, guides the selective

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use of costly imaging, and is particularly valuable where echocardiography and specialist care are scarce (Maganti et al., 2010).

Yet contemporary studies highlight substantial limitations in routine practice. In asymptomatic older adults in primary care, auscultation by experienced general practitioners showed low sensitivity (32-44%) for newly detected valvular heart disease, making it a poor standalone screening tool, especially in overweight patients (Gardezi et al., 2018). Systematic review evidence shows wide variability in diagnostic accuracy and underscores that clinicians often over-rely on auscultation despite inconsistent performance and declining training emphasis (Davidsen et al., 2023). These epidemiological realities and performance gaps reframe the importance of improving cardiac auscultation rather than abandoning it: better teaching methods, digital and electronic stethoscopes, multimodal AI and telehealth-enabled phonocardiography all aim to enhance signal quality and interpretation, so that a fundamentally low-cost bedside skill can contribute more effectively to earlier CVD diagnosis and to reducing the global cardiovascular burden (Barrett et al., 2017).

While deep learning has promised to transform the phonocardiogram (PCG) from a subjective clinical art into an objective data science, the transition from laboratory prototypes to point-of-care deployment is fundamentally bottlenecked by two interconnected failures: algorithmic fragility under non-stationary noise and a severe sim-to-real domain gap. Contemporary literature reveals a pervasive “5-class ceiling” in automated auscultation. Despite achieving high accuracy, state-of-the-art architectures are frequently restricted to identifying broad categories—typically Normal, Aortic Stenosis, Mitral Regurgitation, Mitral Stenosis, and Mitral Valve Prolapse—failing to resolve the complex hemodynamic spectrum required for clinical utility (Fang et al., 2025; Wang et al., 2026; Sarker et al., 2025). Furthermore, models trained on static or idealized noise conditions inherently fail when exposed to hospital ambient interference (Wang et al., 2025; Ali et al., 2023). More critically, beyond additive noise, the field largely overlooks the structural spectral morphology shifts caused by human thoracic acoustics—a non-linear physical gap that purely simulated algorithms cannot bridge.

To systematically decouple and address these limitations, this study introduces an *in silico* proof-of-concept framework designed to investigate adaptation mechanisms for the high-granularity classification of 33 distinct cardiac sounds. We propose a unified ResNet18-MLP architecture (11.5M parameters) deliberately engineered with a dual purpose: a rigid convolutional residual backbone designed to extract noise-invariant features, paired with a dense 280K parameter Multi-Layer Perceptron (MLP) classification head to provide the necessary parametric plasticity for subsequent domain adaptation. In Phase 1, we establish a robust structural baseline using 2,010 samples from a high-fidelity auscultation manikin. By optimizing the backbone from scratch on Mel-spectrograms under extreme dynamic hospital noise augmentation ($U(-5, 5)$ dB SNR), our framework successfully breaks the 5-class limit, achieving an average validation F1-score of ~ 0.813 and an independent SAM test F1-macro score of 73.6%.

Crucially, we hypothesize that purely manikin-based training—while statistically necessary to capture rare pathological sound profiles without massive clinical data collection—inevitably fails when interfacing with real human physiology. In Phase 2, we explicitly quantify this structural domain shift by deploying the Phase 1 model zero-shot on real-world smartphone recordings from healthy human subjects, where its accuracy collapses to 15.7%. We subsequently demonstrate a parameter-efficient domain adaptation strategy via head-only fine-tuning (freezing the backbone and updating only the dense MLP head). By addressing both the taxonomic granularity in simulation and the mathematical realities of adapting to human target domains, this framework establishes a transparent blueprint for off-device acoustic AI. Ultimately, we aim to expose the mechanics of sim-to-real transfer, explicitly cautioning that true clinical deployment remains impossible until these adaptation pathways are rigorously fine-tuned on diverse, pathological human cohorts.

2. Related Work

2.1. Architectural Bottlenecks in Multi-Class Classification

As established, expanding the diagnostic taxonomy beyond binary or limited-class paradigms introduces severe feature entanglement. State-of-the-art models attempt to manage this by scaling architectural complexity, often at the cost of clinical applicability. For instance, while Fang et al. (2025) achieved 95.6% accuracy, their approach relies on complex multi-channel coupling features and ECG synchronization, rendering the method impractical for single-channel digital stethoscopes. Similarly, Wang et al. (2026) utilized Injected Noise Dual Attention Networks (INDANet) to attain high accuracy, yet their feature extraction pipeline remains strictly optimized for standard categories, failing to separate continuous and subtle overlapping murmurs.

Efforts to expand diagnostic horizons have frequently relied on computationally prohibitive architectures. Sarker et al. (2025) proposed a Fuzzy Rank Based Ensemble integrating VGG, EfficientNet, and DenseNet to manage multi-disease overlaps, while Cheng and Sun (2023) deployed CAFusionNet, combining parallel CNN and Transformer branches. Although effective, these heavy architectures (Li et al., 2025; Cheng and Sun, 2023) and MFCC-based pipelines (Hossein-zadeh et al., 2024; Hasan and Karim, 2025) highlight a critical technical gap. We fundamentally question the current trend of defaulting to massive Transformers or complex ensemble models. Instead, we argue that standard residual learning (e.g., ResNet) has been severely underestimated for 2D spectral signals (spectrograms); when combined with the correct data augmentation strategy, residual connections provide the necessary geometric capacity for high-granularity separation without the prohibitive computational overhead.

2.2. Extreme Noise Robustness (< 5 dB SNR)

Clinical environments are characterized by chaotic acoustic interference, often exceeding the amplitude of the cardiac signal itself. Recent studies confirm that standard denoising fails when the Signal-to-Noise Ratio (SNR) approaches 0 dB. Wang et al. (2025) demonstrated that a Feature-Shuffle Multi-Head Attention Autoencoder was required to reconstruct signals at -6 dB SNR, a finding corroborated by the LU-Net architecture which utilized LSTM-based spectral tracking to survive hospital ambient noise (Ali et al., 2023).

The necessity of testing at extreme limits is further supported by work on Phase-Space Boundary Decomposition (PSBD) in pink noise at 0 dB (Hu et al., 2025) and the boundary identification metrics proposed by Torre-Cruz et al. (2022). Hybrid architectures, such as the permutation-coded U-Nets (Aldelemy et al., 2025) and attention-based denoising autoencoders (Wang et al., 2024), confirm that deep learning models must be actively conditioned to ignore multipath artifacts (Zhang et al., 2019). Our dynamic noise augmentation strategy, utilizing a continuous range of $\mathcal{U}(-5, 5)$ dB, directly aligns with this rigorous standard, proving that robust residual architectures can overcome severe masking without specialized denoising modules.

2.3. Sim-to-Real Transfer: The Manikin Paradigm

The reliance on synthetic or simulated data is a contentious but increasingly validated methodology. Torabi et al. (2025) fundamentally altered this landscape with the HLS-CMDS dataset, proving that recordings from the SAM 3G[®] manikin capture the spatial acoustic attenuation necessary for robust training. Helyer et al. (2024) and industry catalogs (SEM Trainers, 2026) further document the SAM 3G[®]'s ability to reproduce 25+ distinct pathological sounds with high physiological fidelity.

Research by Tsai et al. (2020) successfully trained deep autoencoders for blind source separation exclusively on manikin data, while recent investigations into Generative Large Language Multi-Modal Models (GLLMs) confirm that High-Fidelity Manikins (HFM) provide sufficient “grounding” for AI generalization (Hamilton et al., 2024). While recent studies analyzing domain shifts suggest these high-fidelity proxies provide essential feature pre-training (Torabi, 2026), our empirical findings reveal that transitioning from this simulated baseline to uncalibrated real-world smartphone sensors inevitably introduces a severe acoustic domain gap, necessitating targeted adaptation strategies.

2.4. Edge AI and Mobile Deployment

The democratization of cardiac screening requires algorithmic efficiency. Zhao et al. (2023) demonstrated the viability of smartphone-based severe Aortic Stenosis detection using lightweight CNNs with Gradient-based Class Activation Maps (Grad-CAM). However, deploying complex classifiers on resource-constrained hardware remains challenging. Safdar et al. (2025) addressed this with SET-MobileNet, achieving 120ms inference times on microcontrollers, while TYNet (Zhang et al., 2025) utilized aggressive parameter pruning for embedded sensors.

Modern architectural optimizations, such as the Fused-MBConv layers in EfficientNetV2B0 adaptations (Alrabie and Barnawi, 2025) and the depthwise separable convolutions in Lightx3 (Baig et al., 2025), illustrate the trade-off between accuracy and FLOPs. While extreme parameter pruning is a popular trend, we strongly defend that our evaluated architectures—ranging from custom 6.5M parameter CNNs up to the 11.2M parameters of our modified ResNet18—occupy a maximum of approximately 42.7 MB in memory. This footprint remains highly viable for true offline inference on any modern smartphone, successfully maintaining the core argument for accessible mobile health without sacrificing the representational depth required for a 33-class taxonomy or domain adaptation plasticity.

2.5. Hemodynamic Complexity: PDA, PH, and Split S2

Detecting subtle hemodynamic anomalies represents the zenith of acoustic diagnosis. Pulmonary Hypertension (PH), characterized by an accentuated P2, was recently screened with an AUROC of 0.79 by Guo et al. (2025),

validating the detectability of high-frequency valve closures (Gaudio et al., 2023). Cross-domain learning approaches, such as the PH-ELM (Gaudio et al., 2025), further support the feasibility of transferring learned acoustic features to human subjects.

Conversely, Patent Ductus Arteriosus (PDA) remains a volatile challenge; a 2026 systematic review (Long et al., 2026) exposed that current PDA algorithms suffer from overfitting and lack external validation, despite earlier ensemble attempts achieving moderate success (Gómez-Quintana et al., 2021). Finally, the temporal resolution required to identify Split S2 sounds has driven the shift from manual dechirping algorithms (Xu et al., 2000) to end-to-end deep learning classifiers capable of resolving millisecond-scale spectral splits (Jani et al., 2021). Our model’s ability to classify these 33 distinct granularities marks a decisive advancement over current “Normal vs. Abnormal” paradigms.

2.6. Acoustic Domain Shift and Parameter-Efficient Adaptation

While high-fidelity simulators provide the essential taxonomic breadth required to model rare pathologies, deploying these models on biological patients inevitably introduces a severe acoustic domain shift. Variations in tissue attenuation, thoracic anatomy, and uncalibrated sensor coupling (e.g., smartphone microphones) drastically alter the spectral morphology of cardiac signals in the real world. Recent studies have rigorously quantified this sim-to-real gap in acoustic and audio-visual domains, demonstrating how physical world variability severely degrades the generalization of models trained on purely simulated representations Guajardo-Herrera et al. (2025); Chen et al. (2024). In broader audio classification and biomedical signal processing, bridging this gap frequently relies on Domain Adaptation and Transfer Learning paradigms.

However, a critical challenge arises when adapting models to small clinical cohorts: extensive fine-tuning on limited real-world datasets risks catastrophic forgetting of the complex, low-support structural anomalies learned during pre-training Yi et al. (2023). To mitigate this, parameter-efficient adaptation strategies—specifically, freezing the deep convolutional feature extractors and updating exclusively the classification head—have emerged as a theoretically grounded method to recalibrate decision boundaries for target domains without destroying the underlying source representations Anonymous (2025). Notably, in the specific context of clinical audio and heart sound analysis, head-only adaptation has proven empirically superior to full fine-tuning for preventing overfitting in low-resource target distributions Biermann et al. (2025). By explicitly quantifying the zero-shot degradation on human subjects and employing a strict Leave-One-Subject-Out (LOSO) head-only fine-tuning protocol, our framework directly addresses this sim-to-real bottleneck, establishing a pragmatic pathway for clinical translation.

3. Methods

3.1. Phase 1: Simulated Dataset Acquisition and Taxonomy

To establish a robust pre-training baseline and overcome the granularity limitations of public datasets, we generated a custom simulated dataset using the SAM 3G[®] Student Auscultation Manikin. This high-fidelity simulator provides acoustically accurate reproductions of 33 distinct cardiac conditions, including rare valve pathologies often absent in clinical datasets.

Audio was captured using a Samsung Galaxy A54 smartphone positioned directly at the manikin’s mitral auscultation focus (fifth intercostal space, midclavicular line). To naturally induce spatial variance and mimic real-world placement inconsistencies, the smartphone was manually removed and repositioned for every single 10-second recording session. This specific anatomical placement was selected to maximize the acoustic energy and tactile feedback of the fundamental $S1$ and $S2$ heart sounds, simulating a realistic digital stethoscope acquisition in a resource-constrained point-of-care setting. The recordings were digitized at a sampling rate of 22 kHz and subsequently downsampled to 6 kHz to reduce computational overhead. The final Phase 1 pre-training dataset comprises 2,010 samples across 33 classes.

3.2. Signal Preprocessing and Universal Augmentation Pipeline

To ensure robust feature extraction and strictly standardize the acoustic interface between the simulated manikin data and the subsequent real-world human recordings (detailed in Section 3.6), all acoustic data was processed through a strict sequential pipeline prior to network ingestion. First, raw audio signals were resampled to a uniform target rate of 6 kHz to prevent aliasing while preserving the Nyquist limit for all diagnostic cardiac frequencies. Let $x(t)$ denote this resampled audio signal, which is visualized as a waveform and Mel-spectrogram in Figure 1a.

During the training phases, physical acoustic interference was simulated in the time domain by linearly combining the clean signal with random snippets of hospital ambient noise (containing chatter, alarms, and mechanical sounds). To prevent overfitting to a single noise amplitude and force the model to extract robust features across a broad continuum of acoustic interference, the mixing was controlled by a dynamic Signal-to-Noise Ratio (SNR) sampled from a uniform distribution $\mathcal{U}(-5, 5)$ dB. The general SNR is defined in decibels:

$$SNR_{dB} = 10 \log_{10} \left(\frac{P_{signal}}{P_{noise}} \right) \quad (1)$$

where P_{signal} and P_{noise} represent the average power of the clean audio and noise snippet, respectively. Let $x_{noisy}(t)$ represent this combined signal. Figure 1b illustrates the resulting noisy waveform and spectrogram under an extreme lower-bound condition of this dynamic range (e.g., -5 dB SNR). To stabilize signal energy fluctuations after noise injection, standard z-score normalization was applied to the resulting waveform:

$$x_{norm}(t) = \frac{x_{noisy}(t) - \mu}{\sigma} \quad (2)$$

where μ and σ are the mean and standard deviation of the signal amplitude.

The normalized time-domain signal was then transformed into the time-frequency domain using a Mel-spectrogram, mapping the linear frequency scale to the non-linear Mel scale to approximate human auditory perception:

$$S_{mel}(f, t) = \log(M \cdot |STFT(x_{norm}(t))|^2) \quad (3)$$

where M is the Mel filter bank matrix comprising 128 bands, and $STFT$ denotes the Short-Time Fourier Transform computed with a Hanning window of length 2048 and hop length 512.

Given that pathological cardiac acoustic energy is predominantly concentrated at lower frequencies (< 1 kHz), we applied a deterministic frequency crop to the generated spectrograms, retaining only the lower 40 Mel bins. This spatial cropping, visible in panel (b) of Figure 1, acts as a hard low-pass filter against high-frequency environmental artifacts and substantially reduces the feature space dimensionality. Finally, instead of standard min-max scaling, the cropped Mel-spectrogram tensors underwent a sample-wise standard scaling (Z-score normalization). This independent standardization ensures that each individual 2D spectrogram possesses a mean of 0 and a variance of 1. In the context of acoustic processing, this sample-wise approach is critical: it effectively neutralizes any residual variations in recording volume or microphone distance without relying on population-level statistics, thereby strictly preventing data leakage between the training and testing partitions prior to convolutional feature extraction.

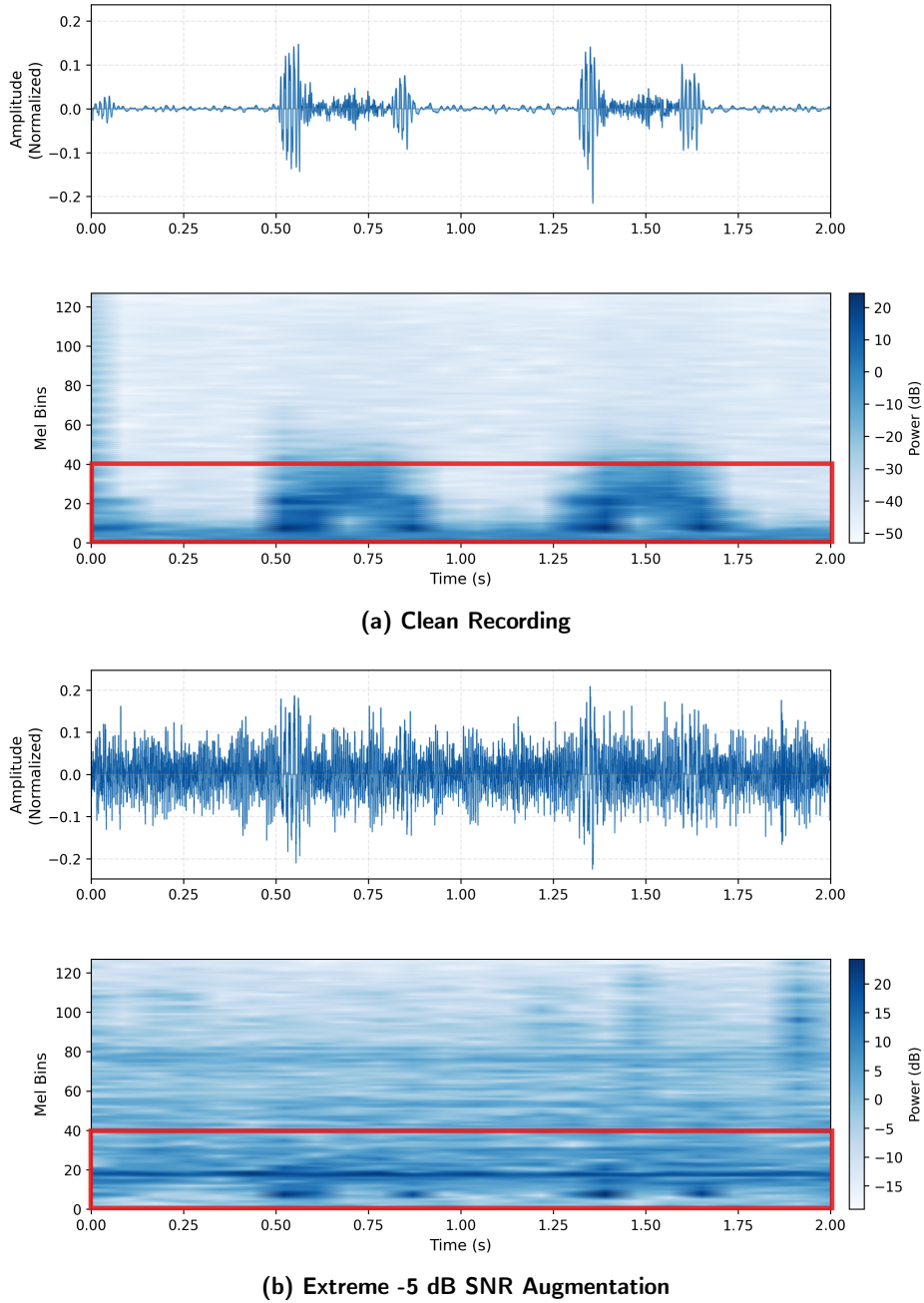


Figure 1: Time-domain waveform and corresponding Mel-spectrogram of a pathological Fourth Heart Sound recorded from the SAM 3G[®] manikin via a digital stethoscope ($f_s = 6$ kHz). Spectrograms visualize the 128-band Mel scale, with a red marker indicating the 40-bin crop used for network input. Panel (a) shows the clean high-fidelity signal. Panel (b) shows the result of injecting hospital ambient noise at a -5 dB SNR condition (representing the lower bound of our dynamic $\mathcal{U}(-5, 5)$ dB training range), demonstrating severe spectral masking of continuous features while the primary rhythmic S1/S2 bursts remain discernable.

3.3. Dynamic Extreme Noise Augmentation

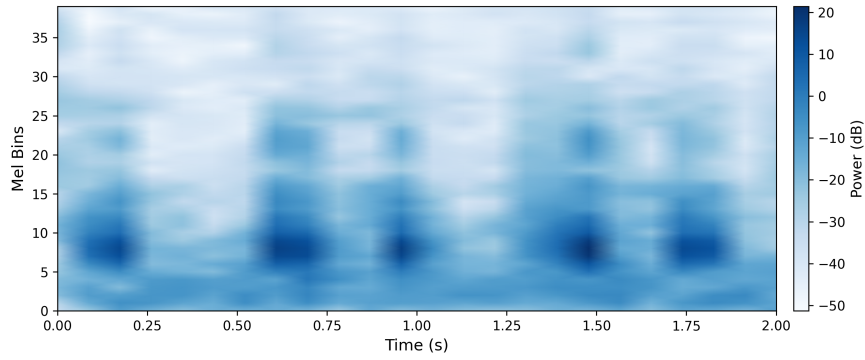
Unlike standard augmentation strategies that employ benign SNR ranges (e.g., 5–15 dB) or static thresholds, we subjected our Phase 1 pre-training pipeline to an extreme, continuous dynamic noise distribution:

$$SNR_{train} \sim \mathcal{U}(-5, 5) \text{ dB} \quad (4)$$

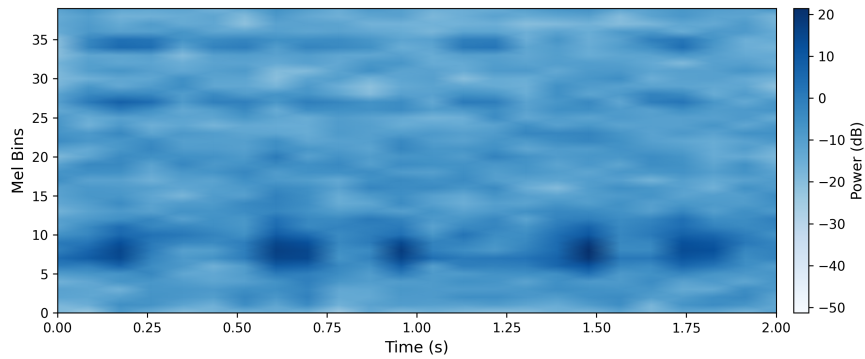
While binary classification tasks (Normal vs. Abnormal) may survive isolated -6 dB SNR environments (Wang et al., 2025), resolving 33 distinct classes, particularly those defined by subtle, high-frequency continuous murmurs like Patent Ductus Arteriosus (PDA), encounters severe acoustic masking. Clinically, hospital ambient noise is profoundly stochastic; fixing a static noise limit mathematically induces overfitting to a single interference amplitude. By training across a uniform distribution ranging from +5 dB down to an extreme -5 dB, the model is forced to learn the true continuous distribution of hospital noise. This mathematical approach prevents the convolutional filters from merely memorizing a specific noise floor, optimizing the residual connections to dynamically isolate structural cardiac anomalies regardless of additive ambient intensity.

As visually evidenced in Figure 2, the progressive acoustic degradation highlights this challenge. While the high-amplitude fundamental rhythmic bursts (S1 and S2) remain partially visible at sub-zero SNRs, the continuous, low-amplitude pathological murmurs fall entirely below the noise floor. Training across the $\mathcal{U}(-5, 5)$ dB spectrum mathematically ensures that the network learns robust global rhythmic features at the lower bounds (-5 dB) while still resolving the subtle spectral textures at the upper bounds (+5 dB) that are strictly necessary for a 33-class taxonomy.

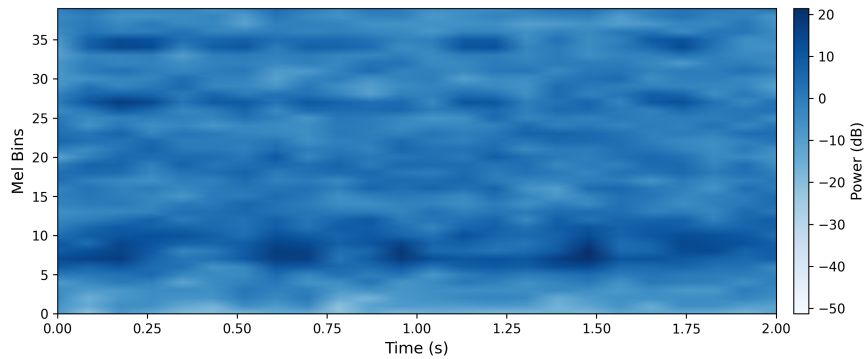
Crucially, we must acknowledge the fundamental limitation of this additive augmentation strategy. While $\mathcal{U}(-5, 5)$ dB noise perfectly simulates environmental hospital interference, it is strictly a linear superimposition. It fundamentally fails to capture the non-linear skin-coupling artifacts, thoracic resonance alterations, and spatial tissue attenuation inherent to human patients. As visualized in Figure 3, the structural spectral morphology of human recordings differs drastically from noise-augmented simulator data. This unmodeled physical variance constitutes the core "sim-to-real gap" that necessitates the Phase 2 domain adaptation protocol.



(a) Clean Signal



(b) 5 dB SNR (Upper Bound of Training Distribution)



(c) -5 dB SNR (Extreme Lower Bound)

Figure 2: Acoustic degradation of a continuous murmur (Patent Ductus Arteriosus) under progressive hospital noise augmentation, visualized with a fixed power scale (dB) to demonstrate relative masking. In the clean signal (a), the continuous pathological texture between the S1/S2 bursts is clearly isolated. At 5 dB (b), representing the upper bound of our noise augmentation ($P_{\text{signal}} > P_{\text{noise}}$), the high-frequency continuous murmur is partially masked but retains clear structural contours. At the extreme lower bound of our dynamic training distribution (-5 dB) (c), the pathological texture is completely obliterated beneath the ambient interference, leaving only the primary rhythmic bursts. Training across this entire continuum ($\mathcal{U}(-5, 5)$ dB) prevents amplitude overfitting and forces the network to learn robust, noise-invariant representations.

Comparación Visual: MelSpectrograms Train vs People

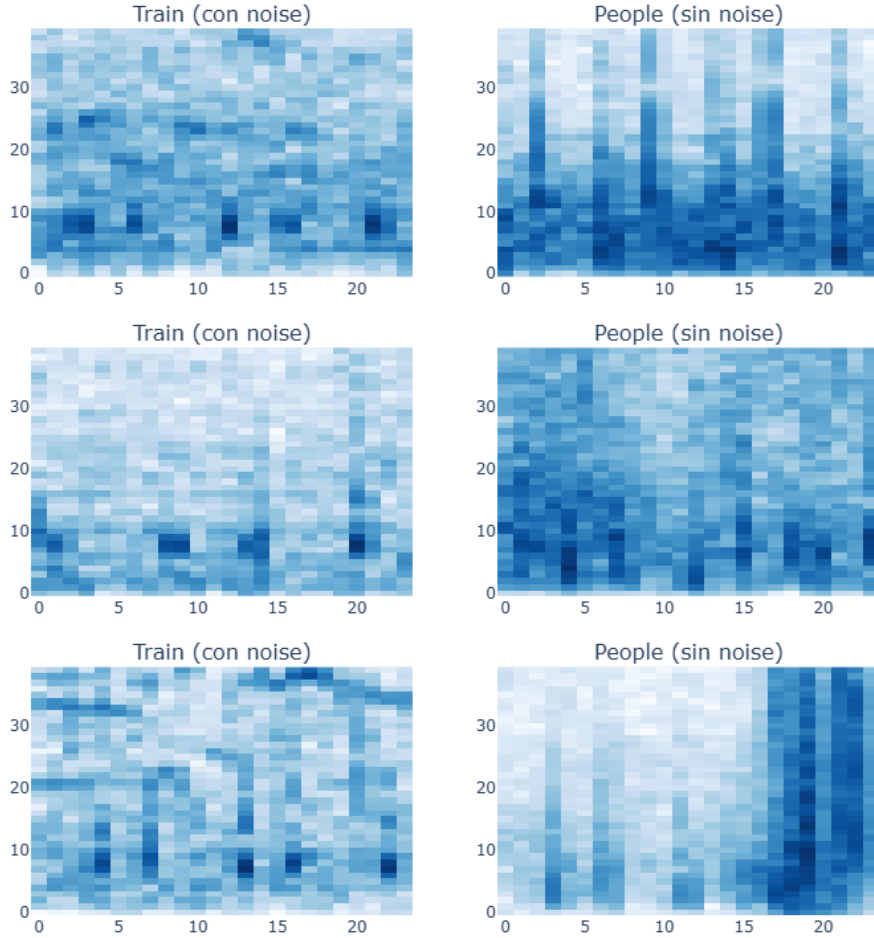


Figure 3: Visual Comparison: Mel-spectrograms from the Phase 1 simulated training set (left, with additive noise) versus real human baseline recordings (right, clean). Despite heavy additive noise in simulation, the structural morphology—specifically the frequency spread of $S1/S2$ and inter-beat harmonic resonance—differs significantly from the real-world human acoustic signature. This confirms that simulated environmental noise cannot act as a proxy for the structural domain shift caused by human tissue.

3.4. Phase 1: Unified Architecture Design (ResNet18-MLP)

To resolve the competing requirements of extreme noise resilience and cross-domain adaptability, the primary classification engine anchoring our framework is a unified, custom-adapted ResNet18-MLP architecture (He et al., 2016). This model, comprising approximately 11.5 million parameters, was deliberately engineered to decouple feature extraction from domain adaptation through a two-part structural strategy.

First, the backbone utilizes classical residual connections to maintain gradient flow and feature representation under extreme dynamic acoustic masking. While originally designed for 3-channel visual tasks, we specifically adapted the network's input stem to process 2D single-channel Mel-spectrogram tensors. Crucially, to prevent premature early spatial downsampling that would destroy the fine-grained frequency resolution of our 40-bin Mel crop, the initial convolution's stride was reduced from 2 to 1, and the subsequent standard max-pooling layer was entirely bypassed using an Identity mapping.

Second, rather than employing the standard single linear classification layer typical of ResNet architectures, we replaced the network's terminus with a dense Multi-Layer Perceptron (MLP) head comprising approximately 280,000 trainable parameters. This specific architectural modification was explicitly designed for Phase 2: while the deep residual backbone provides the geometric rigidity necessary to extract structural features under Phase 1 noise, the dense MLP head provides the "parametric plasticity" required to recalibrate decision boundaries and absorb structural domain shifts during subsequent head-only fine-tuning.

To benchmark this architectural choice against current edge-deployment trends, the ResNet18 backbone was evaluated alongside established lightweight architectures (MobileNetV2 (Sandler et al., 2018) and EfficientNetV2 (Tan and Le, 2021)) under a strict pairwise comparison of two initialization paradigms: Transfer Learning utilizing standard ImageNet pre-trained weights, and initialization strictly from scratch.

3.5. Phase 1: Pre-Training Protocol and Validation Strategy

To rigorously preclude data leakage and establish a robust Phase 1 baseline, the 2,010-sample simulated dataset was partitioned using a strict session-wise split rather than a random temporal frame split. Crucially, the independent hold-out test set ($N = 401$) was constructed exclusively from physically independent recording sessions captured on separate days. Because the smartphone sensor was manually repositioned between every capture, the test set inherently evaluates the model's resilience to subtle spatial attenuation and sensor coupling shifts, guaranteeing that the network learns generalized pathological acoustic signatures rather than memorizing the continuous ambient noise floor of a single prolonged recording. Phase 1 model selection and hyperparameter tuning were conducted using a 5-fold Stratified Cross-Validation strategy on the simulated training subset, guaranteeing proportional class representation across all data splits.

A critical hypothesis tested within our protocol was the efficacy of initialization paradigms under extreme acoustic stress. While transfer learning is a standard practice, we hypothesized that training entirely *from scratch* would yield superior results. For acoustic tasks subject to dynamic ambient noise, discarding visual pre-trained weights forces the convolutional filters to optimize exclusively for complex spectral morphologies and continuous temporal rhythms, rather than relying on geometric edge detectors that may fail to transfer under heavy acoustic masking.

To guarantee a strictly fair comparison while accommodating these distinct initialization strategies, optimization hyperparameters were carefully tailored. Models trained from scratch were optimized using the Adam optimizer with an initial learning rate of 1×10^{-3} , an L_2 weight decay of 1×10^{-2} , and a *ReduceLROnPlateau* reduction factor of 0.8. Conversely, to prevent catastrophic forgetting of pre-trained visual representations, the transfer learning configurations utilized a more conservative learning rate of 1×10^{-4} , a weight decay of 1×10^{-3} , and a reduction factor of 0.5.

All architectures employed a class-weighted Cross-Entropy loss function to directly mitigate the inherent class imbalance of the dataset. Training was capped at 1000 epochs with a batch size of 32, tightly constrained by an Early Stopping callback monitoring the validation loss with a patience of 20 epochs. Finally, the optimal weights for each fold were explicitly selected by monitoring the maximum macro F1-score, ensuring the final evaluated models were highly resilient to minority classes rather than simply minimizing global loss.

3.6. Phase 2: Sim-to-Real Domain Adaptation Protocol

To explicitly address the structural domain gap identified upon deploying the Phase 1 models to the real world, we designed a proof-of-concept Domain Adaptation protocol utilizing human recordings. Our clinical adaptation cohort comprised 11 independent smartphone recordings collected from 7 human subjects (5 female, 2 male; ages 25–61). To rigorously test the model's robustness against real-world physiological variance, the cohort included subjects with

pre-existing conditions (obesity, medically managed hypertension, and arthritis) alongside healthy baselines. All real-world audio was segmented and processed using the exact universal preprocessing pipeline established in Section 3.2.

Given the severely limited size of this real-world target distribution, standard full-network fine-tuning would inevitably result in massive catastrophic forgetting of the 33-class simulated taxonomy. Therefore, we employed a parameter-efficient *head-only* fine-tuning strategy leveraging our unified architecture. The deep residual backbone of the ResNet18-MLP (optimized in Phase 1 to extract noise-invariant spectral morphologies) was strictly frozen. Only the dense 280,000-parameter MLP classification head was updated. This isolates the parametric plasticity, allowing the model to recalibrate its decision boundaries for human tissue attenuation without overwriting the core geometric feature extractors.

The adaptation dataset combined the real-world human chunks with a proportionally balanced subset of the original simulated training data. Because the human cohort consisted of healthy physiological rhythms, all real-world human chunks were assigned the “Normal 75BPM” target label. We explicitly acknowledge that training a flexible MLP head exclusively toward a single target class inherently risks a mathematical expansion of the “Normal” decision boundary—a phenomenon quantified in our Phase 2 results.

To strictly preclude subject-level data leakage, validation was conducted using a 7-fold Leave-One-Subject-Out (LOSO) cross-validation scheme. In each fold, all recordings from a single subject were held out for evaluation, while the remaining 6 subjects were used for fine-tuning. Models were explicitly re-initialized from their original Phase 1 pre-trained checkpoints at the start of each fold to prevent progressive weight leakage. Optimization utilized the Adam optimizer (learning rate 1×10^{-4} , weight decay 1×10^{-4}) and unweighted Cross-Entropy loss over 15 epochs, modulated by a Cosine Annealing scheduler. After validating generalization via LOSO, a final proof-of-concept model was exported by training on the complete 7-subject cohort.

4. Results

4.1. Phase 1: Simulated Classification Performance

To ensure strict methodological rigor and prevent data leakage, the simulated dataset was partitioned using a strict *recording-wise* split. This guarantees that temporal acoustic segments originating from the same continuous manikin recording are never shared between training and testing subsets. This methodology successfully isolated an independent hold-out test set of 401 samples (approximately 20% of the corpus) strictly for final Phase 1 evaluation.

Evaluated under the extreme dynamic noise condition ($\mathcal{U}(-5, 5)$ dB SNR), the unified ResNet18-MLP architecture demonstrated highly stable feature extraction capabilities. During the 5-fold Stratified Cross-Validation on the training subset, the model achieved an average validation F1-score of **~0.813**. When deployed to the completely unseen independent hold-out test set, the model maintained a resilient macro-averaged F1-score of **73.6%**. This performance confirms that the deep residual backbone successfully extracts generalized pathological acoustic signatures across the 33-class taxonomy, rather than merely memorizing the continuous ambient noise floor or overfitting to specific acoustic domains.

While the extreme $\mathcal{U}(-5, 5)$ dB masking inevitably degrades absolute deterministic accuracy for highly volatile continuous murmurs (such as Patent Ductus Arteriosus) that fall entirely below the sub-zero noise floor, the architecture functions successfully as a robust differential diagnostic safety net. The residual connections preserve the structural contours of these complex hemodynamic anomalies within the immediate differential horizon (the highest probability logits). This empirically confirms that even when the network struggles to isolate the absolute primary diagnosis under severe environmental noise, the underlying structural anomaly is captured, establishing a mathematically rigorous, noise-invariant baseline prior to introducing real-world domain shifts.

4.2. Phase 1: Architectural Search and Backbone Justification

To empirically justify the structural components of our unified ResNet18-MLP architecture, we conducted a preliminary baseline evaluation against a broader experimental space. This search evaluated established mobile-optimized architectures (MobileNetV2 and EfficientNetV2) and a grid of custom lightweight CNNs under identical dynamic noise conditions.

The generalization gap between these exploratory architectures within the simulated domain is visually captured in Figure 4. The scatter plots illustrate a critical trade-off between model size and validation-to-test performance drop under severe masking. While custom CNNs (e.g., 4CNN-1MLP) exhibited high validation recall—benefiting from the parametric plasticity of their dense MLP heads—their steeper vertical drop on the independent test set indicated

Table 1

Preliminary performance metrics of experimental baseline architectures evaluated under dynamic noise augmentation. The inclusion of the pre-trained ResNet18 highlights the severe performance degradation caused by visual Transfer Learning, ultimately justifying the train-from-scratch residual backbone of our unified model.

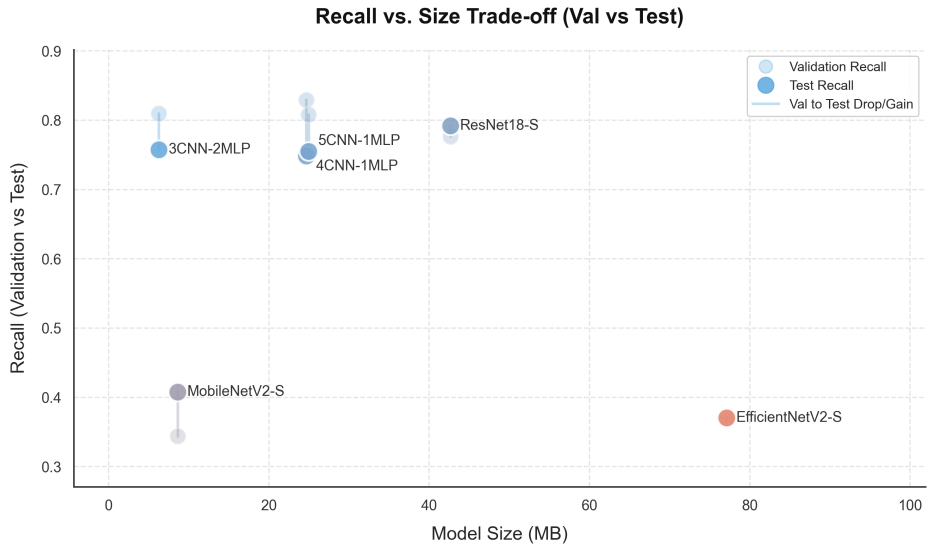
Architecture	Top-1 Acc	Top-3 Acc	Macro F1	Parameters	Size (MB)
EfficientNetV2-S (Scratch)	36.16%	64.94%	35.71%	20.2M	77.0
MobileNetV2 (Scratch)	49.63%	65.07%	40.94%	2.3M	8.6
ResNet18 (Pre-trained)	56.31%	77.07%	48.53%	11.2M	42.7
Custom 4-Block CNN	79.55%	92.00%	75.20%	6.5M	24.7
ResNet18 Backbone (Scratch)	84.04%	93.77%	79.97%	11.2M	42.7

a severe vulnerability to the simulated dynamic noise distribution. Conversely, architectures utilizing deep residual connections demonstrated exceptional geometric stability, maintaining their performance metrics when transitioning to unseen noisy data.

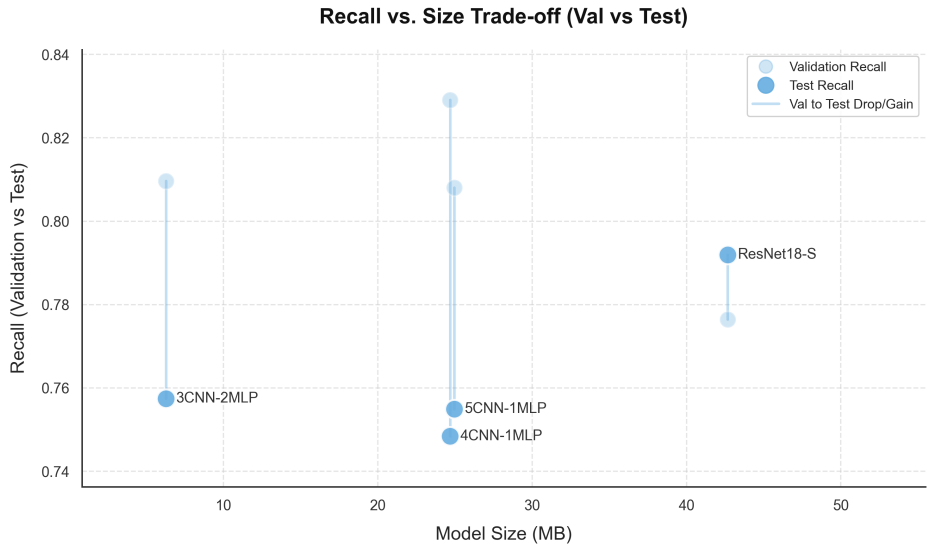
This finding directly informed our unified design: the custom CNNs proved that while dense classification heads are highly plastic, they require a deep, rigid convolutional backbone to extract noise-invariant features. Fusing these concepts yielded our final ResNet18-MLP.

A critical empirical finding from this baseline comparison was the systemic failure of visual Transfer Learning. As quantified in Table 1, initializing architectures with standard ImageNet weights uniformly degraded performance when subjected to dynamic acoustic masking. Most notably, a standard ResNet18 initialized from scratch achieved an 84.04% Top-1 accuracy in this preliminary evaluation, whereas its pre-trained counterpart collapsed to a 56.31% Top-1 accuracy. This performance delta confirms that forcing 2D Mel-spectrograms through convolutional filters previously optimized for 3-channel visual edge detection is fundamentally counterproductive. Discarding visual priors allows the network to optimize exclusively for continuous temporal rhythms and subtle spectral morphologies.

Ultimately, architectures conventionally optimized for visual efficiency experienced severe performance degradation. MobileNetV2 yielded a Top-1 accuracy of only 49.63%, and EfficientNetV2 collapsed entirely to 36.16%. This provides empirical evidence that while depthwise separable convolutions are computationally efficient, they lack the representational robustness required to isolate overlapping cardiac spectral features from high-density hospital noise. The standard ResNet18 from scratch provided the optimal topological rigidity for noise extraction, which strictly rationalizes its selection as the backbone for our finalized 11.5M parameter ResNet18-MLP framework.



(a) Global Experimental Space



(b) High-Performance Architectures

Figure 4: Recall vs. Model Size trade-off illustrating the generalization gap between validation (light blue) and independent test performance (dark blue) during our preliminary architectural search. Custom CNNs and MobileNetV2 exhibited a significant performance collapse (steep vertical drop) under dynamic noise. The superior resilience of residual learning justified the selection of the ResNet backbone for our unified architecture.

4.3. Phase 1: Class-wise Sensitivity and Top-k Analysis

Table 2 details the exhaustive performance metrics for all 33 diagnostic categories evaluated using the unified ResNet18-MLP architecture. The framework demonstrated remarkable diagnostic fault tolerance within the simulated domain, successfully capturing the true underlying pathology within its Top-3 predictions (100% Top-3 accuracy) in 15 out of the 33 classes, despite the severe $\mathcal{U}(-5, 5)$ dB dynamic noise masking.

Notable clinical findings from this pre-training baseline include:

- **High Robustness in Structural Anomalies:** Severe structural and valvular pathologies such as *Acute Pericarditis*, *Mitral Stenosis*, *Tetralogy of Fallot*, and *Congenital Aortic Stenosis* achieved a 100% Top-1 accuracy. This indicates that their spectral signatures maintain a resilient structural integrity that easily survives extreme ambient masking without defaulting to a generic rhythm.
- **The Top-3 Differential "Safety Net":** The most compelling validation of the residual backbone emerged in the acoustically volatile classes. *Patent Ductus Arteriosus (PDA)*, a subtle continuous murmur entirely obliterated by ambient noise, recorded a 0.00% Top-1 accuracy due to complete spectral masking. However, its Top-3 accuracy surged to 66.67%. Similarly, complex overlapping hemodynamics like *Aortic Stenosis Regurgitation* saw an accuracy spike from 60.00% (Top-1) to 100.00% (Top-3). This confirms that the model accurately preserves these elusive conditions as high-probability differential candidates, even when absolute deterministic prediction fails under severe noise.
- **Physiological Distinction:** The model successfully isolated *Physiological Split S2* with a high 86.96% accuracy, distinguishing it from pathological splits (e.g., Atrial Septal Defects, which suffered lower accuracy due to noise overlap). Establishing this capability in simulation represents a critical prerequisite for avoiding false positives and unwarranted anxiety in primary care and pediatric screening.

4.4. Phase 1: Error Analysis and the Limits of Acoustic Masking

While the dynamic noise augmentation effectively regularized the network globally within the simulated domain, an analysis of the class-wise performance highlights the strict physical limitations of acoustic masking. As detailed in our error analysis, highly subtle or continuous hemodynamic anomalies suffered substantial degradation under the $\mathcal{U}(-5, 5)$ dB training distribution.

Notably, Patent Ductus Arteriosus (PDA) and Pulmonary Hypertension exhibited severe drops in Top-1 recall. PDA, characterized by a continuous machine-like murmur, is entirely enveloped by the ambient noise floor at lower SNR bounds, causing the network to misclassify it as a base rhythm. However, the residual architecture still extracts sufficient secondary features to push the correct pathology into the highest probability outputs, with PDA jumping from a Top-1 recall of 0.0000 to a Top-3 accuracy of 0.6667. Conversely, high-frequency valve closures like Pulmonary Hypertension suffer near-total masking under extreme sub-zero SNR (0.0000 Top-1, 0.1667 Top-3), forcing the network to default almost entirely to an underlying rhythm. Crucially, understanding these specific simulated failure modes—where severe acoustic masking forces the network to default to a 'Normal' base rhythm—provides the essential comparative baseline for analyzing the reciprocal domain-shift errors encountered when deploying this model zero-shot to uncalibrated human subjects in Phase 2.

Table 2

Detailed performance metrics by class for the ResNet18-MLP architecture on the independent **simulated** test set ($N = 401$). Precision, Recall (Top-1 Accuracy), and F1-Score are presented alongside Top-3 Accuracy. "Top-3 Gain" represents the accuracy improvement when considering the top three probability logits. Classes are sorted descending by F1-Score to highlight robust spectral signatures.

Class	Support	Precision	Recall (Top-1)	F1-Score	Top-3 Acc	Top-3 Gain
Acute Pericarditis	6	1.0000	1.0000	1.0000	1.0000	+0.0000
Aortic Regurgitation Acute	5	1.0000	1.0000	1.0000	1.0000	+0.0000
Congenital Aortic Stenosis	6	1.0000	1.0000	1.0000	1.0000	+0.0000
Mitral Stenosis and Regurgitation	21	1.0000	1.0000	1.0000	1.0000	+0.0000
Normal 110BPM	20	1.0000	1.0000	1.0000	1.0000	+0.0000
Pulmonary Stenosis	6	1.0000	1.0000	1.0000	1.0000	+0.0000
Third Heart Sound	22	0.9565	1.0000	0.9778	1.0000	+0.0000
Mitral Stenosis	20	0.9524	1.0000	0.9756	1.0000	+0.0000
Normal 90BPM	20	0.9524	1.0000	0.9756	1.0000	+0.0000
Summation Sound	21	0.9091	0.9524	0.9302	0.9524	+0.0000
Normal 60BPM	22	1.0000	0.8636	0.9268	0.9545	+0.0909
Aorta Coarctation	6	1.0000	0.8333	0.9091	0.8333	+0.0000
Venous Hum	6	1.0000	0.8333	0.9091	1.0000	+0.1667
Physiological Split S2	23	0.9091	0.8696	0.8889	0.9130	+0.0435
Tetralogy of Fallot	6	0.7500	1.0000	0.8571	1.0000	+0.0000
Aortic Stenosis Severe	6	0.8333	0.8333	0.8333	0.8333	+0.0000
Congestive Heart Failure	6	0.8333	0.8333	0.8333	0.8333	+0.0000
Aortic Regurgitation	6	1.0000	0.6667	0.8000	1.0000	+0.3333
Mitral Regurgitation Mild	20	0.7391	0.8500	0.7907	0.9000	+0.0500
Mitral Valve Prolapse	21	0.8824	0.7143	0.7895	0.9048	+0.1905
Fourth Heart Sound	30	0.8462	0.7333	0.7857	0.9667	+0.2333
Atrial Septal Defect	6	0.7143	0.8333	0.7692	0.8333	+0.0000
Aortic Stenosis Severe S2 Absent	6	0.8000	0.6667	0.7273	0.8333	+0.1667
Innocent Systolic Murmur Vibratory	6	0.5455	1.0000	0.7059	1.0000	+0.0000
Normal 75BPM	21	0.5429	0.9048	0.6786	0.9524	+0.0476
Aortic Stenosis Regurgitation	5	0.7500	0.6000	0.6667	1.0000	+0.4000
Atrial Fibrillation	6	0.5556	0.8333	0.6667	1.0000	+0.1667
Mitral Regurgitation Marked	22	0.5161	0.7273	0.6038	0.9545	+0.2273
Hypertrophic Cardiomyopathy	6	1.0000	0.1667	0.2857	0.5000	+0.3333
Innocent Systolic Murmur	6	0.0000	0.0000	0.0000	0.3333	+0.3333
Patent Ductus Arteriosus (PDA)	6	0.0000	0.0000	0.0000	0.6667	+0.6667
Pulmonary Hypertension	6	0.0000	0.0000	0.0000	0.1667	+0.1667
Ventricular Septal Defect (VSD)	6	0.0000	0.0000	0.0000	0.5000	+0.5000
Overall (Weighted Avg)	401	0.8093	0.8180	0.8018	0.9177	+0.0997

Table 3

Error Analysis: Top Misclassifications under Dynamic $\mathcal{U}(-5,5)$ dB SNR Augmentation for the unified ResNet18-MLP model. The error counts reflect the exact number of misclassified samples relative to the class support in the test set. The data empirically reveals that continuous and subtle murmurs are frequently masked by extreme negative SNR noise, predominantly resulting in misclassification as underlying Normal rhythms.

Actual Pathology	Primary Misclassification	Error Count	Acoustic / Clinical Implication
Patent Ductus Arteriosus (PDA)	Normal 75BPM	6 / 6	Continuous murmur masked by sub-zero SNR noise
Pulmonary Hypertension	Normal 75BPM	6 / 6	Accentuated P2 obscured, defaulting to base rhythm
Hypertrophic Cardiomyopathy	Innocent Systolic Murmur	5 / 6	Systolic ejection murmurs present acoustic overlap
Innocent Systolic Murmur	Mitral Regurgitation Mild	4 / 6	Shared mid-systolic spectral frequencies
Ventricular Septal Defect (VSD)	Congestive Heart Failure	2 / 6	Holosystolic murmur overlap with failure artifacts
Venous Hum	Fourth Heart Sound	1 / 6	Low-frequency continuous hum confused with S4
Fourth Heart Sound	Mitral Regurgitation Marked	9 / 30	Pre-systolic low frequencies blurring into systole

4.5. Phase 2: Sim-to-Real Domain Gap & Parameter-Efficient Fine-Tuning

To explicitly quantify the hypothesized sim-to-real acoustic domain gap, the Phase 1 pre-trained ResNet18-MLP model was deployed zero-shot onto the uncalibrated real-world human cohort ($N = 7$ healthy subjects, 11 total recordings). The empirical results revealed a severe generalization collapse. Under zero-shot conditions, the model correctly predicted a "Normal" rhythm in only 15.7% of the human acoustic segments.

Analysis of the probability logits and intermediate feature representations indicated that the network was not failing randomly, but rather experiencing a systematic structural confusion. The attenuated high-frequency features and altered spatial resonance of the human thorax caused the mean real-world Mel-spectrograms to closely align with the simulated decision boundaries of highly specific structural anomalies. Mathematically, the human baseline records mimicked

Table 4

Phase 2 Domain Adaptation Results for the unified ResNet18-MLP architecture. Comparison of zero-shot (Original) versus head-only fine-tuned (Adapted) performance on the real-world human cohort, alongside the corresponding retention of simulated validation metrics.

Model Configuration	Human 'Normal' Accuracy (%)	SAM F1-Macro (%)
ResNet18-MLP (Phase 1 Original / Zero-Shot)	15.7%	73.6%
ResNet18-MLP (Phase 2 Adapted)	98.6%	69.9%

Hypertrophic Cardiomyopathy (cosine similarity = 0.9404) and *Mitral Regurgitation* (cosine similarity = 0.9185). This catastrophic zero-shot failure definitively confirms that high-fidelity purely simulated training cannot blindly generalize to human patients. The visual proof of this spectral overlap is illustrated in Figure 5.

To mitigate this, we executed the head-only fine-tuning protocol. By freezing the robust spectral feature extractors learned in Phase 1 and updating exclusively the 280,000-parameter multi-layer perceptron (MLP) head, the architecture demonstrated remarkable domain adaptation plasticity. Evaluated under the strict 7-fold Leave-One-Subject-Out (LOSO) cross-validation scheme, the human Normal detection rate surged. The final model, trained on the complete 7-subject adaptation cohort, recovered the baseline physiological rhythm with an exceptional 98.6% accuracy.

However, establishing true scientific rigor requires acknowledging the algorithmic trade-offs inherent in few-shot clinical fine-tuning. As visualized in Figure 6 and quantified in Table 4, forcing the dense MLP head to accommodate the human domain resulted in a measurable degradation of the original 33-class simulated taxonomy. The independent SAM F1-macro score dropped from 73.6% to 69.9%. This establishes a highly successful proof-of-concept for adapting heavily masked acoustic representations, but simultaneously highlights the mathematical cost of boundary recalibration.

4.6. Phase 2: Catastrophic Forgetting and Sacrificed Classes

While the parameter-efficient adaptation successfully bridged the sim-to-real gap for healthy baseline rhythms, establishing true scientific rigor requires acknowledging the algorithmic trade-offs inherent in few-shot clinical fine-tuning. As detailed in Table 4, forcing the dense MLP head to accommodate the human domain resulted in a partial degradation (catastrophic forgetting) of the original 33-class simulated taxonomy. Specifically, the independent SAM Test F1-macro score dropped from 73.6% down to 69.9% for the unified ResNet18-MLP model.

A granular paired sample analysis of the Phase 2 predictions revealed the specific acoustic cost of this adaptation. To successfully classify the muffled, heavily attenuated human recordings as "Normal" and achieve the 98.6% accuracy, the network systematically expanded its geometric decision boundary for the base rhythm. Consequently, acoustically adjacent and subtle simulated pathologies were erroneously absorbed into this expanded "Normal" class. Most notably, conditions such as *Mitral Regurgitation*, *Mitral Valve Prolapse (MVP)*, and *Fourth Heart Sound (S4)* experienced severe metric degradation post-adaptation.

Because the Phase 2 training signal aggressively biased the classification head toward identifying human-specific spatial attenuation as a healthy "Normal 75BPM", conditions characterized by subtle mid-systolic overlaps (MVP, MR) or pre-systolic low-frequency textures (S4) were mathematically overwritten by the expanded boundary. We transparently report this degradation to validate that the 98.6% recovery on healthy humans is a targeted recalibration, not a flawless diagnostic victory. This emphasizes that while head-only fine-tuning effectively maps a model to human baselines, the model's ability to distinguish real-world pathological recordings from healthy ones remains fundamentally unvalidated. True clinical deployment mandates future work involving extensive fine-tuning on pathological clinical cohorts to reconstruct these complex pathological boundaries in the real world.

5. Discussion

To contextualize our findings within the current landscape of automated auscultation, we summarize the primary contributions of this work before detailing their clinical and technical implications:

- **High-Granularity & Edge Deployment:** We break the 5-class ceiling by delivering a 73.6% independent test F1-macro score across 33 classes using a unified ResNet18-MLP architecture. With a footprint of 11.5M parameters, the framework remains strictly optimized for true offline smartphone inference, contrasting with the

heavy computational requirements of recent Transformer and ensemble methods (Sarker et al., 2025; Cheng and Sun, 2023).

- **Sim-to-Real Validation & Adaptation:** Leveraging the validated acoustic fidelity of the SAM 3G[®] manikin (Torabi et al., 2025; Helyer et al., 2024), we utilize a balanced corpus of 2,010 high-fidelity samples to isolate acoustic variables (Phase 1). Crucially, we move beyond simulation by explicitly quantifying the severe sim-to-real structural domain gap on uncalibrated human subjects (a collapse to 15.7% zero-shot accuracy) and demonstrating a successful parameter-efficient fine-tuning mitigation strategy (Phase 2). We transparently validate this 98.6% adaptation recovery as a mathematical expansion of the "Normal" decision boundary, defining the inherent trade-offs of sim-to-real transfer.
- **Dynamic Noise Resilience:** Adopting an aggressive “resilience-based” training strategy, we augment data across a continuous dynamic range of $\mathcal{U}(-5, 5)$ dB SNR to harden the residual backbone. This mirrors the sub-zero noise tolerance emerging as the rigorous new gold standard in biomedical signal processing (Torre-Cruz et al., 2022; Wang et al., 2024).

5.1. The Residual Advantage in Extreme Acoustic Variance

Our empirical comparison during the initial architectural search revealed a critical insight regarding acoustic feature extraction: extremely lightweight networks inherently lack the geometric capacity to model the variance introduced by a dynamic noise range of -5 to 5 dB. When hospital ambient interference forces the SNR below zero, standard depthwise separable convolutions or simple pooling layers tend to discard heavily masked signals as noise, leading to performance collapse. Conversely, the identity shortcuts within the ResNet18 backbone allowed gradients to flow effectively, preserving the structural contours of acoustic features despite severe masking. However, our Phase 2 findings add a critical corollary: while the residual backbone dominates in additive noise rigidity, it requires a dense classification head to exhibit the parametric plasticity necessary for bridging structural domain shifts. This fundamental trade-off directly dictated the design of our unified ResNet18-MLP architecture.

5.2. Breaking the “5-Class Ceiling”

Current state-of-the-art models for heart sound classification typically achieve high accuracy only by restricting the problem space to binary outcomes or coarse 5-class taxonomies. Our results challenge this paradigm. By achieving a 73.6% independent test F1-macro score across 33 distinct categories under stochastic noise, we demonstrate that properly regularized deep learning models can capture subtle spectral differences—such as the distinction between a *Vibratory Innocent Murmur* and a *Pathological Systolic Murmur*—that are fundamentally lost in lower-granularity studies.

5.3. The Value of Top-3 Accuracy in Telemedicine

The discrepancy between deterministic Top-1 predictions and broader Top-3 inclusion rates under extreme noise is not a failure, but a robust feature of clinical utility. In a telemedicine context, a “black box” prediction of a single disease can be dangerous, especially when ambient hospital noise forces the SNR below 0 dB. A system that successfully captures volatile murmurs within its top three probability logits—such as PDA surging from 0.00% (Top-1) to 66.67% (Top-3) despite complete spectral masking—serves as a highly reliable differential diagnostic safety net. It effectively narrows the spectrum of cardiac pathology, prompting the clinician to investigate a focused set of differentials rather than starting from a blank slate.

5.4. Robustness vs. Reality: Quantifying and Mitigating the Domain Gap

Rather than a mere limitation, utilizing the SAM 3G[®] manikin serves as a fundamental step of methodological control. Collecting an almost balanced, human-derived dataset of 33 specific pathologies with identical recording hardware is epidemiologically prohibitive. The high-fidelity simulator isolates acoustic variables, providing a controlled *ground truth* to stress-test the models against extreme noise prior to introducing the uncontrollable anatomical variance of human subjects.

While the dynamic $\mathcal{U}(-5, 5)$ dB SNR augmentation successfully enforces robust feature learning, our error analysis (Table 3) reveals the strict acoustic threshold of this extreme regularization. Subtle and continuous hemodynamic anomalies suffer from severe acoustic masking at the lower bounds of our noise distribution. Under these sub-zero

SNR conditions, the network defaults to identifying the surviving fundamental S1 and S2 energetic bursts, which explains the misclassification of these specific pathologies as the underlying base rhythm in simulation.

However, the most significant revelation of this study is the empirical quantification of the reciprocal sim-to-real structural domain gap. When deployed zero-shot to human subjects in Phase 2, the pre-trained model experienced a catastrophic performance collapse (dropping to 15.7% accuracy for normal rhythms). As proven by our cosine similarity analysis, this was not random error, but a deterministic mathematical failure: human tissue attenuation fundamentally altered the spectral resonance, causing the network to confidently mistake healthy human rhythms for *Hypertrophic Cardiomyopathy* and *Mitral Regurgitation*. This definitively proves that high-fidelity simulation cannot blindly generalize to human thoracic acoustics.

Addressing this critical bottleneck, we demonstrate that fine-tuning only the dense MLP classification head with a small number of unlabeled real-world recordings can significantly map the network to human spatial attenuation. By recovering human baseline accuracy to 98.6%, we establish a viable, computationally efficient proof-of-concept pathway for sim-to-real transfer. Crucially, we scientifically agree that this high adaptation accuracy is fundamentally a mathematical expansion of the “Normal” decision boundary. By transparently reporting the corresponding drop in the simulated F1-macro score (to 69.9%) and the swallowing of adjacent classes, we establish that clinical adaptation is a strict computational trade-off, rather than a flawless diagnostic fix.

MelSpectrogram Promedio: Clases de Interés vs People

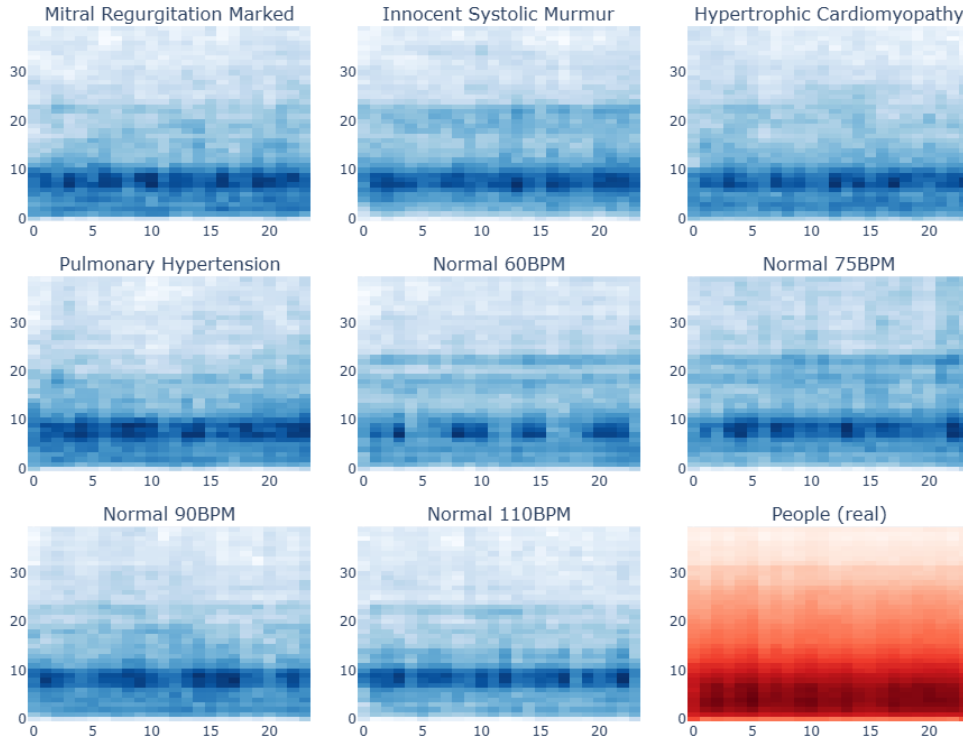


Figure 5: Visual Proof of Zero-Shot Failure (Cosine Similarity Overlap). A comparison of mean Mel-spectrograms for simulated pathologies (top two rows) versus real-world healthy people (bottom right). Due to human tissue attenuation, the "People (real)" spectral signature mathematically mimics the spatial resonance of *Hypertrophic Cardiomyopathy* (0.9404) and *Mitral Regurgitation* (0.9185), causing the 15.7% zero-shot classification collapse.

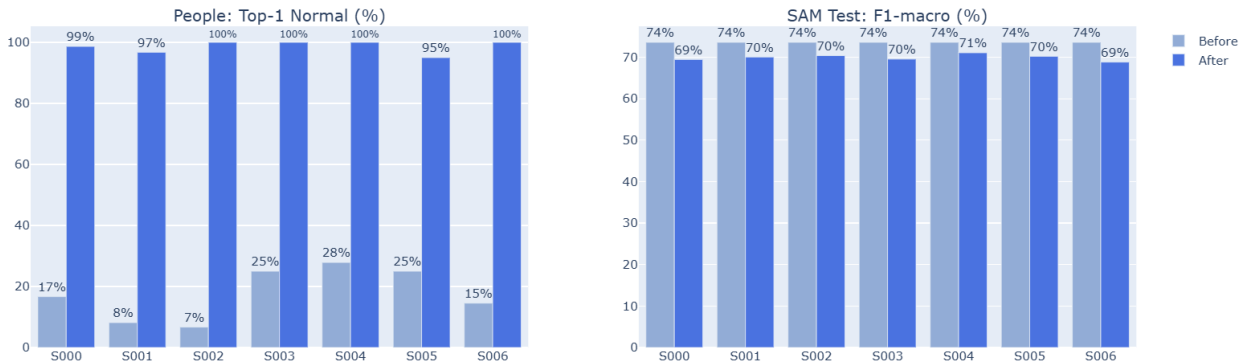


Figure 6: Phase 2 Domain Adaptation Trade-off. The left panel shows the dramatic recovery of the "People" (Normal) class accuracy, jumping from 15.7% (zero-shot) to 98.6% (head-only fine-tuning) across individual subjects. The right panel illustrates the subsequent mathematical cost: the model experiences a controlled degradation in simulated diagnostic performance, with the SAM Test F1-macro score dropping from 73.6% down to 69.9% as the decision boundaries recalibrate.

5.5. The Efficacy of Data-Centric AI for Edge Deployment

By framing this challenge as a *Data-Centric* rather than exclusively a *Model-Centric* problem, we demonstrate that meticulous data curation and targeted adaptation extract optimal performance from established architectures without requiring overly complex ensembles. Our comparative analysis reveals that mobile-optimized architectures like MobileNetV2 (Sandler et al., 2018) and EfficientNetV2 (Tan and Le, 2021) collapse under sub-zero SNR conditions, indicating that depthwise separable convolutions are structurally too “fragile” to isolate cardiac features under heavy spectral masking.

This finding decisively contrasts with the current trend in the literature, where high-granularity results are increasingly pursued through massive Transformer-based networks (Cheng and Sun, 2023; Sarker et al., 2025) at a prohibitive computational cost. By utilizing classical residual connections trained entirely from scratch under a $\mathcal{U}(-5, 5)$ dB noise distribution, our framework achieves high-granularity taxonomic resolution without such overhead. Occupying a maximum memory footprint of just 42.7 MB (11.5M parameters), our unified ResNet18-MLP fits effortlessly within the RAM constraints of standard mobile devices. This proves that high-performance acoustic AI does not inherently require edge-incompatible models, making true offline inference entirely viable for smartphone-based screening in remote and resource-constrained environments.

6. Conclusion

This study presents a two-phase, data-centric methodological framework for high-granularity cardiac auscultation using smartphone-grade audio. By shifting the focus from architectural over-design to rigorous dataset curation and extreme dynamic noise augmentation ($\mathcal{U}(-5, 5)$ dB SNR), we demonstrated in Phase 1 that a unified ResNet18-MLP architecture successfully classifies 33 distinct simulated cardiac conditions. Achieving an independent test F1-macro score of 73.6% under severe stochastic masking, the pre-trained residual backbone provides a highly resilient differential diagnostic safety net in controlled acoustic environments.

Crucially, Phase 2 of this study empirically quantified a severe sim-to-real structural domain gap. When deploying this simulated baseline zero-shot to uncalibrated human subjects, the accuracy collapsed to 15.7% due to human thoracic acoustics mathematically mimicking pathological spectral signatures. By executing a parameter-efficient, head-only fine-tuning protocol on a small real-world cohort, we successfully mitigated this domain shift, recovering the normal physiological baseline detection to 98.6%. However, we transparently quantified the mathematical cost of this adaptation: expanding the “Normal” decision boundary to accommodate human tissue attenuation inevitably swallowed adjacent simulated pathologies, degrading the simulated F1-macro score to 69.9%.

While our framework establishes a robust proof-of-concept for adapting high-fidelity simulated models to real-world acoustic profiles, critical limitations remain. Methodologically, the real-world adaptation cohort relied on assuming the recordings from the 7 human subjects as a generic “Normal 75BPM” class without strict electrocardiographic validation. Furthermore, because our human cohort consisted solely of healthy baseline rhythms, the expanded decision boundary’s ability to accurately distinguish real pathological recordings from healthy ones remains entirely unvalidated. Addressing these boundaries through extensive clinical datasets encompassing diverse, real-world pathologies is an essential prerequisite before evaluating the system’s true clinical efficacy in improving patient triage and outcomes.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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CRedit authorship contribution statement

Efraín-Ismael Flores-Hernández: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data curation, Writing - original draft, Visualization, Project administration. **Maria G. Zavala-Cerna:** Validation, Writing - review & editing.

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